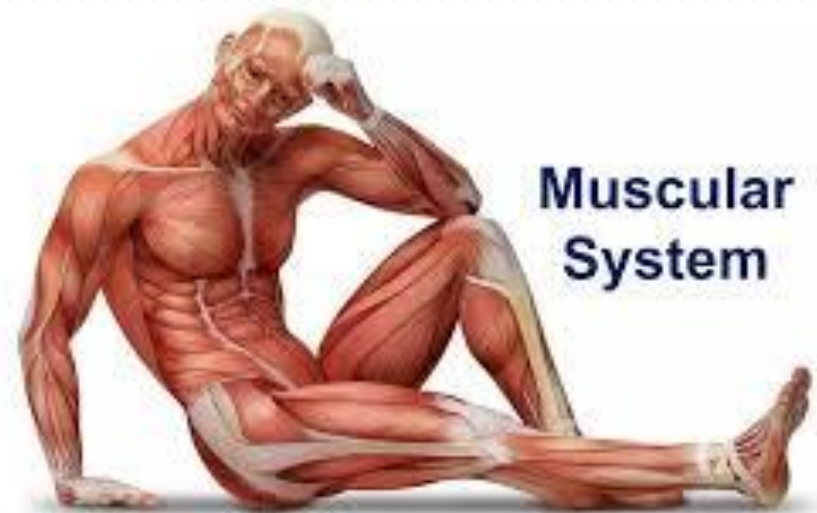


Management of patient with Musculoskeletal diseases



**Muscular
System**

Dr . Bardies Shaban



الحديث الجانبي يشتت الانتباه



من فضلك اغلق الهاتف

2- Hip Replacement

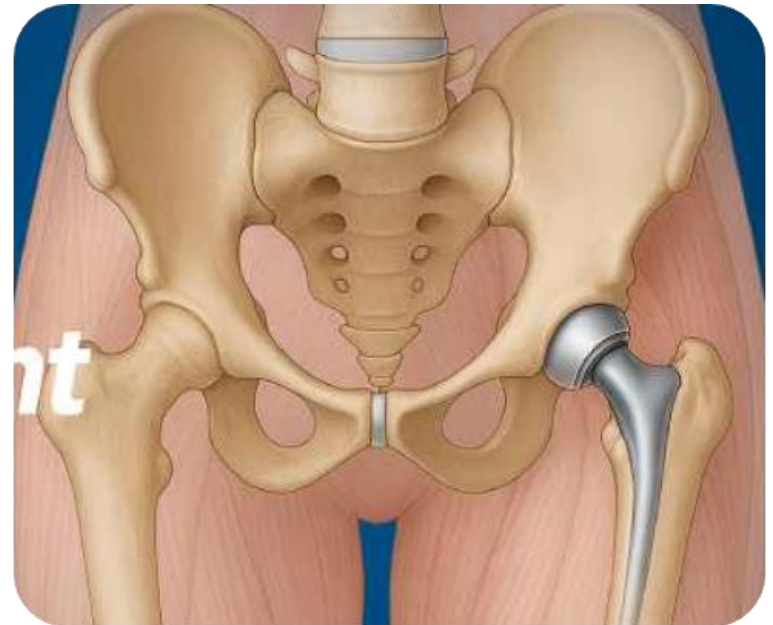
- ◉ Definition
- ◉ Causes
- ◉ Types
- ◉ Manifestations
- ◉ Managements
- ◉ Complications

•Definition:

Surgery to remove damaged sections of the hip joint and replace them. Replacement parts are usually made of metal, ceramic and hard plastic. Also called total hip arthroplasty.



- ❖ Hip replacement might be an option if hip pain **interferes with daily activities** and **nonsurgical** treatments haven't helped and not effective.



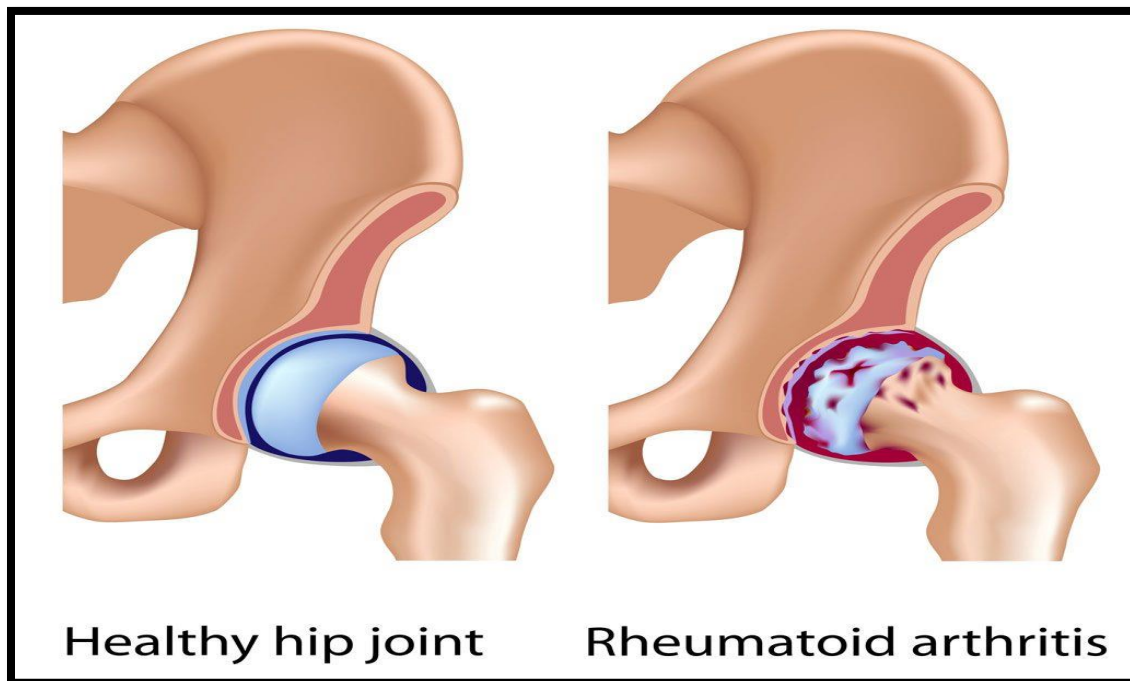
Reasons for replacement:

- **Osteoarthritis**

Most common cause

- **Rheumatoid Arthritis**

Autoimmune condition that causes chronic joint inflammation

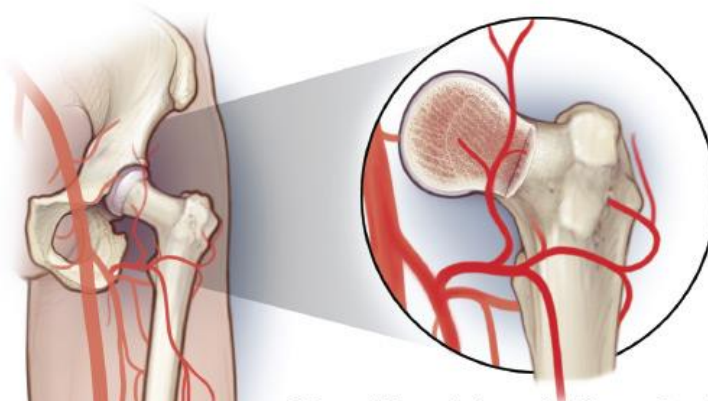


Reasons for replacement:

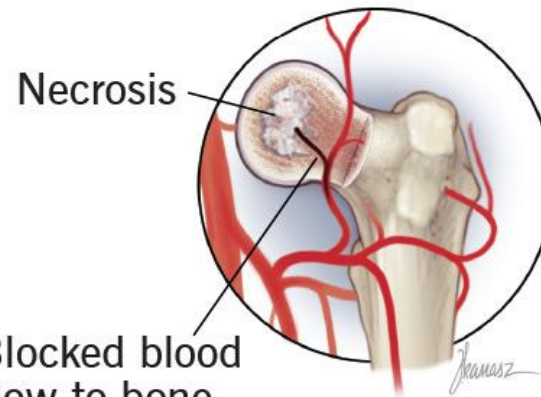
- **Hip Fracture:** common in older adults after falls
- **Avascular Necrosis**

Bone tissue death due to poor blood supply (can result from alcohol, steroids, or trauma)

Avascular necrosis (AVN or osteonecrosis)



Healthy blood flow to bone

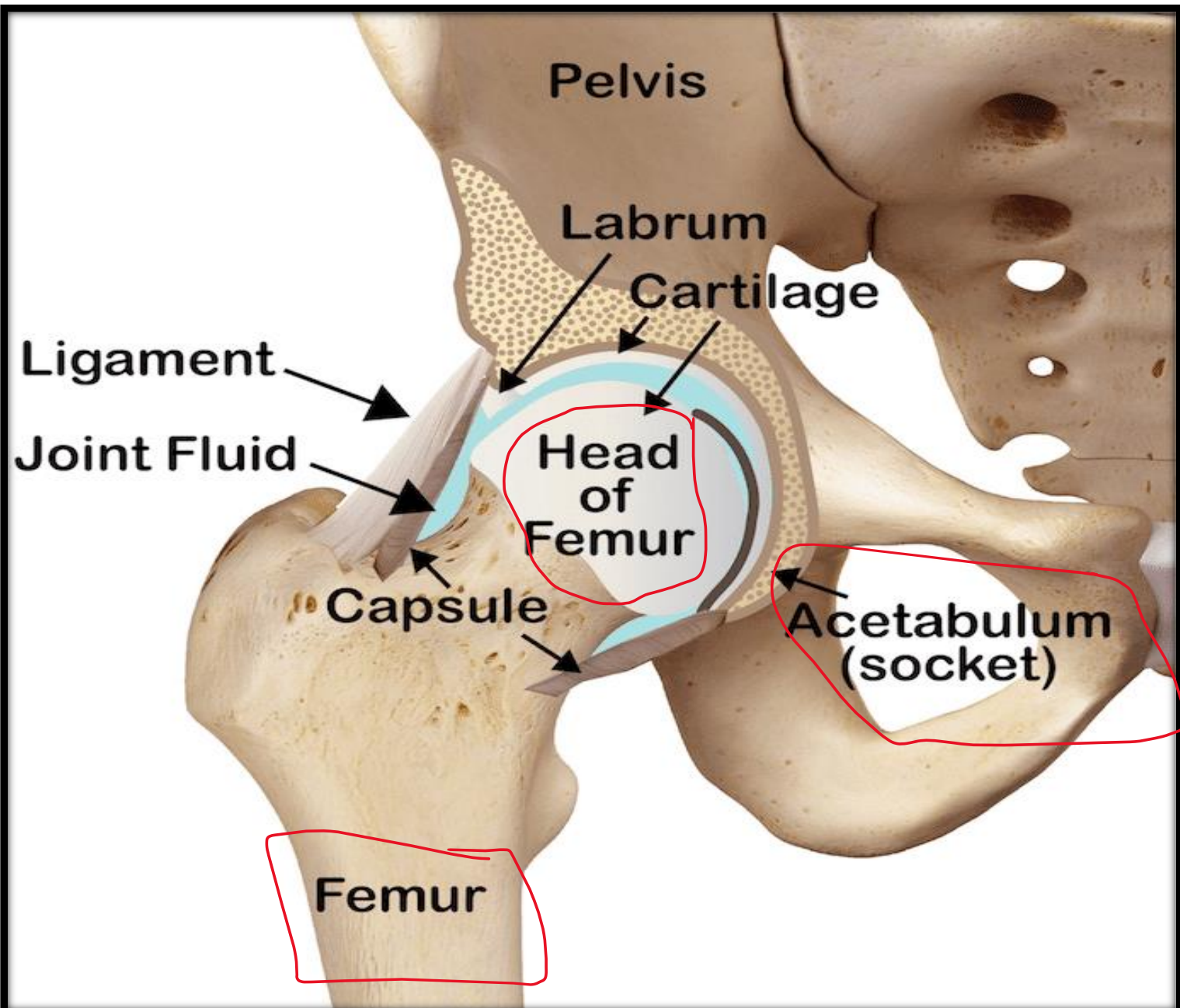


Types of hip replacement

Total hip replacement

Partial hip replacement

Hip Resurfacing



Hip bone
(pelvis)

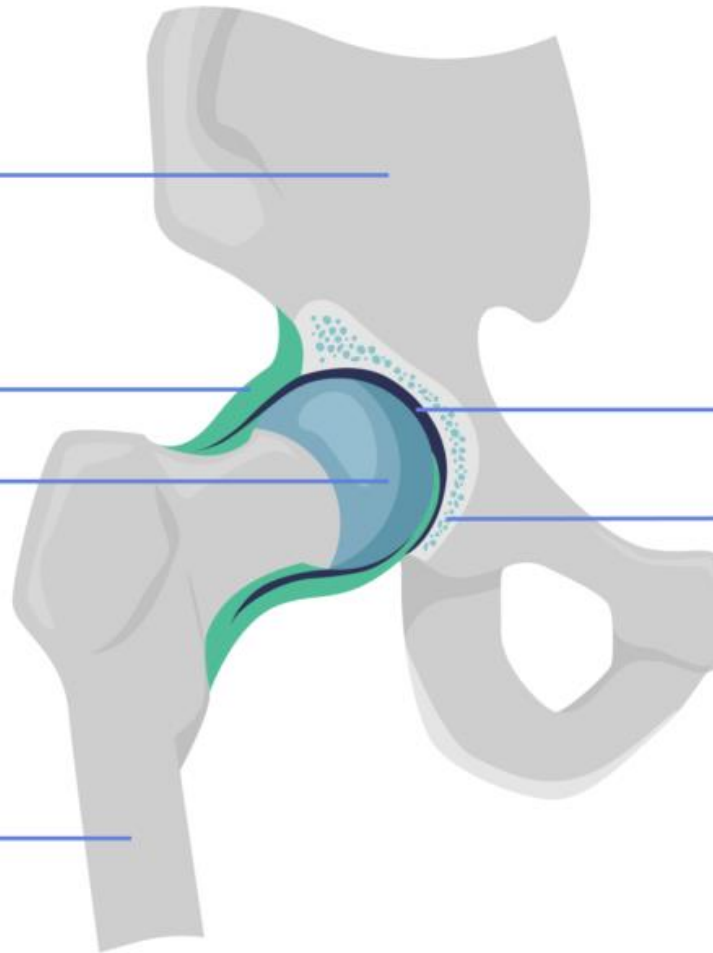
Joint capsule

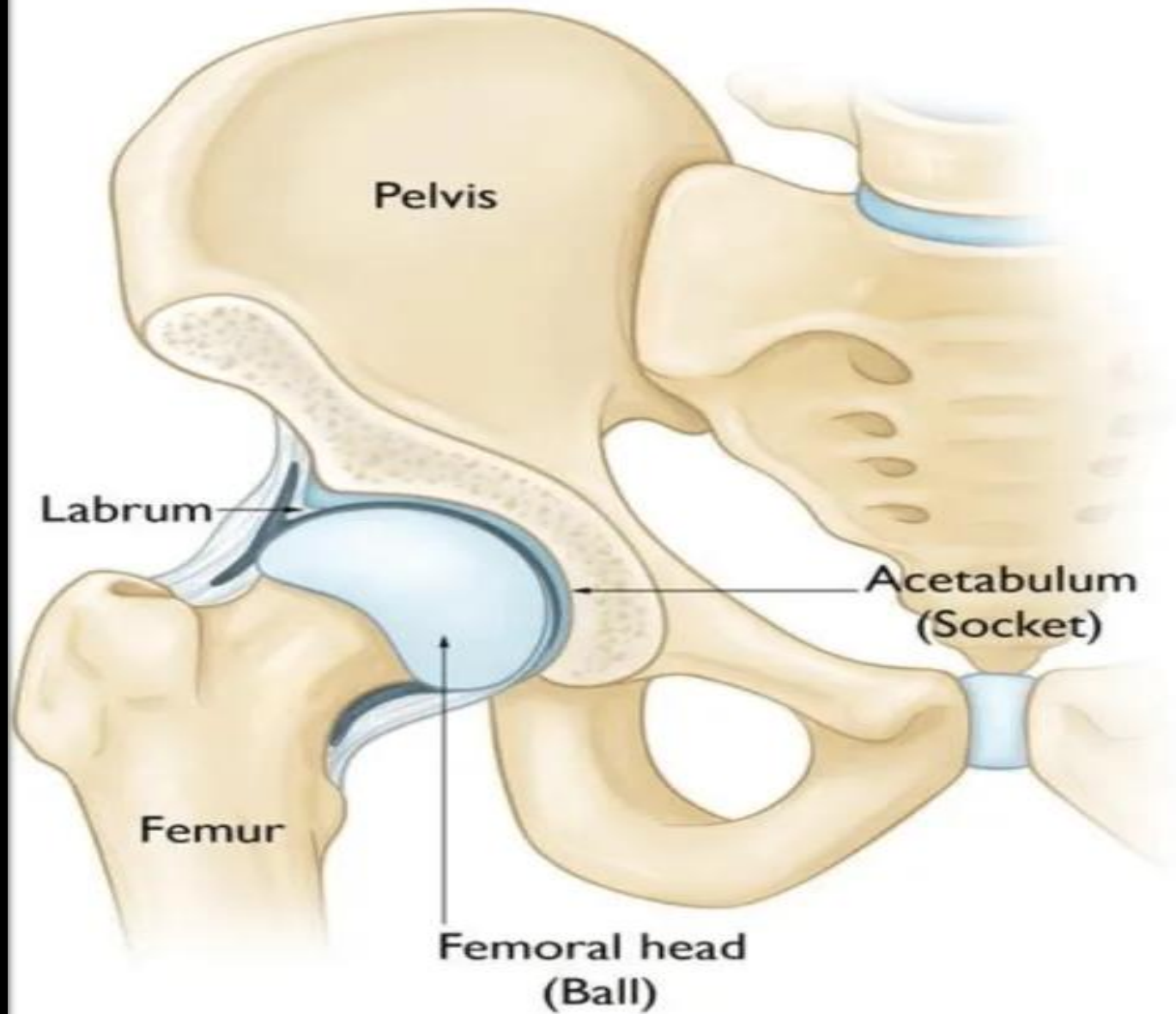
Femoral
head

Thigh bone
(femur)

Cartilage lining

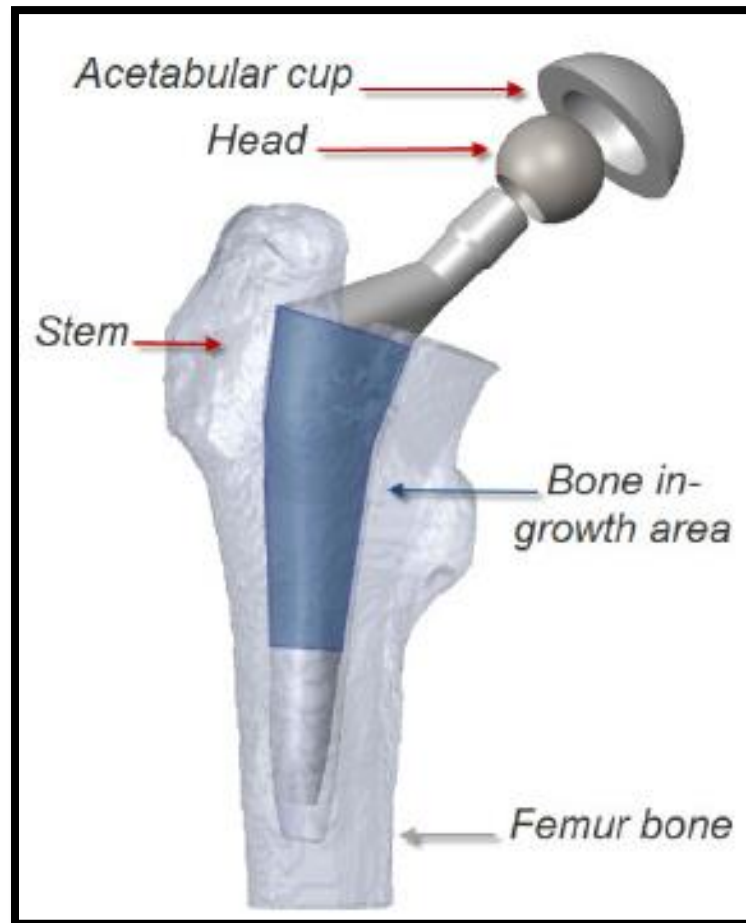
Socket
(acetabulum)

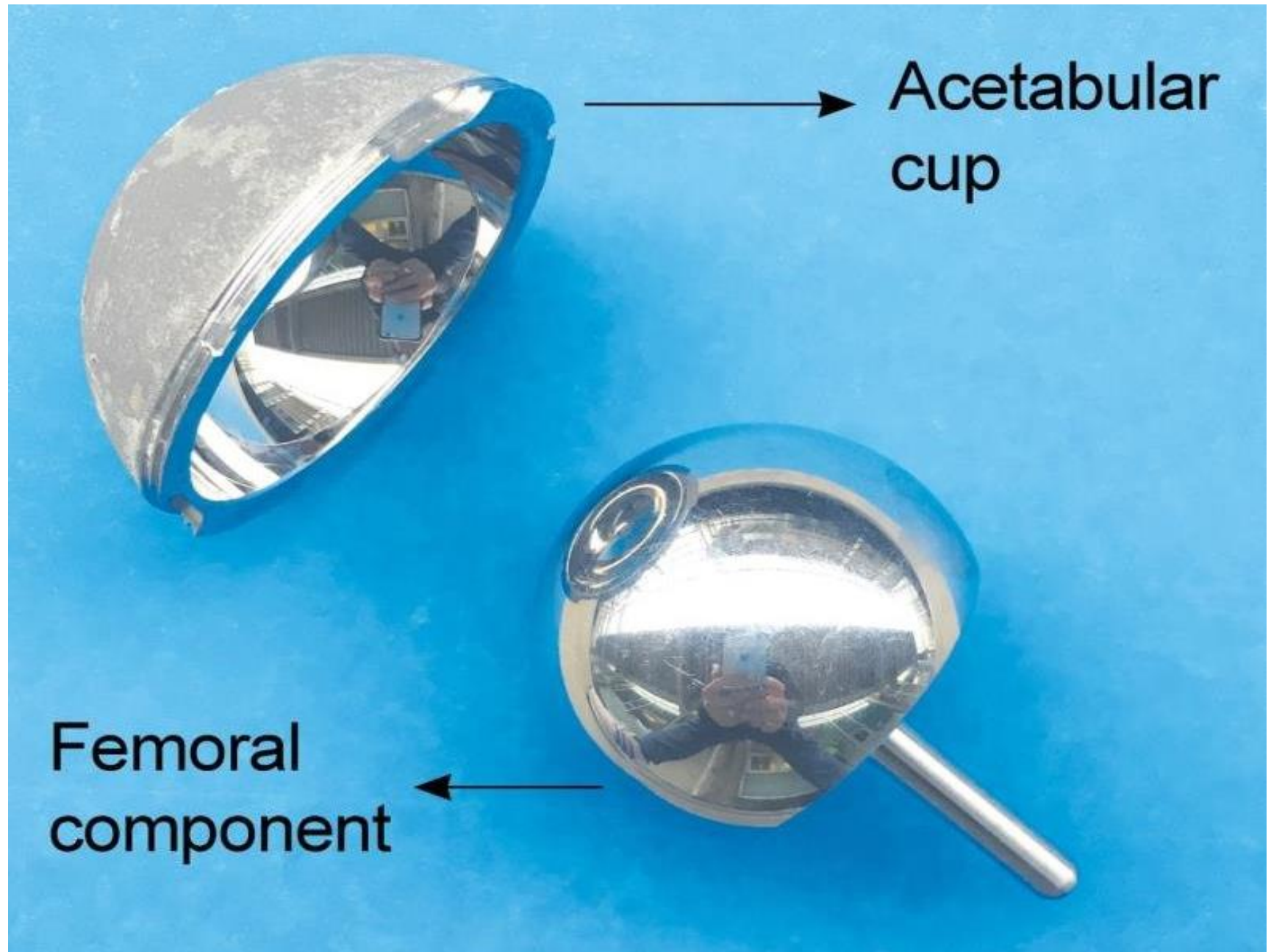




1- Total hip replacement (THR)

- o Most common
- o Replaces : **Acetabulum** (socket), and **Femoral head** (ball)



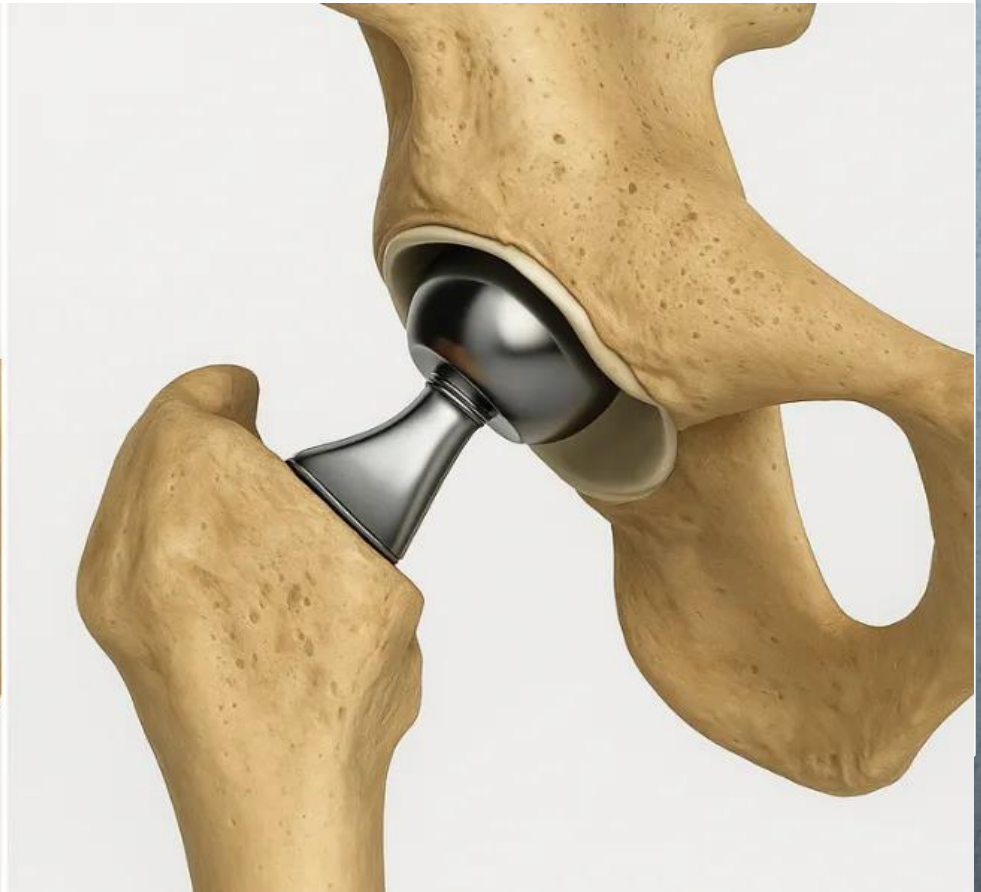


Acetabular
cup

Femoral
component

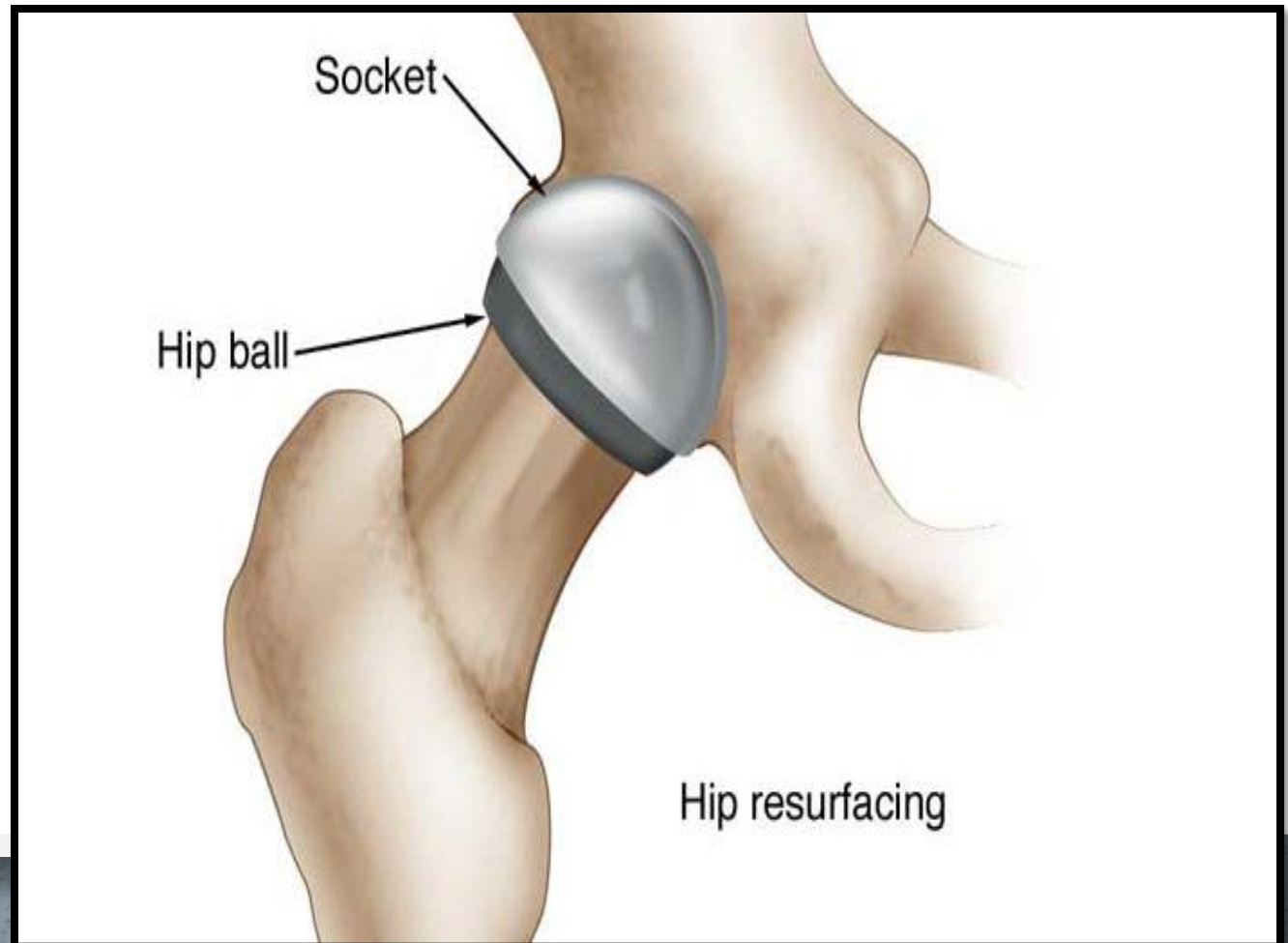
2- Partial hip replacement (PHR)

- Only the **femoral head** is replaced



3- Hip Resurfacing

- Metal cap is placed over the femoral head instead of removing it
- More common in younger patients.



TYPES OF HIP REPLACEMENT SURGERY

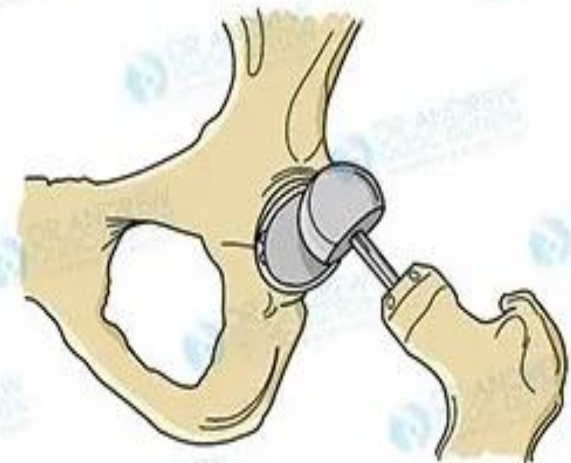
Total Hip Replacement
(total hip arthroplasty)



Partial Hip Replacement
(hemiarthroplasty)



Hip Resurfacing



Nursing process for patients Undergoing Orthopedic Surgery.

1- Assessment

- Vital signs
- Level of consciousness
- Wound drainage
- Breath sounds
- Bowel sounds
- Fluid balance
- Pain



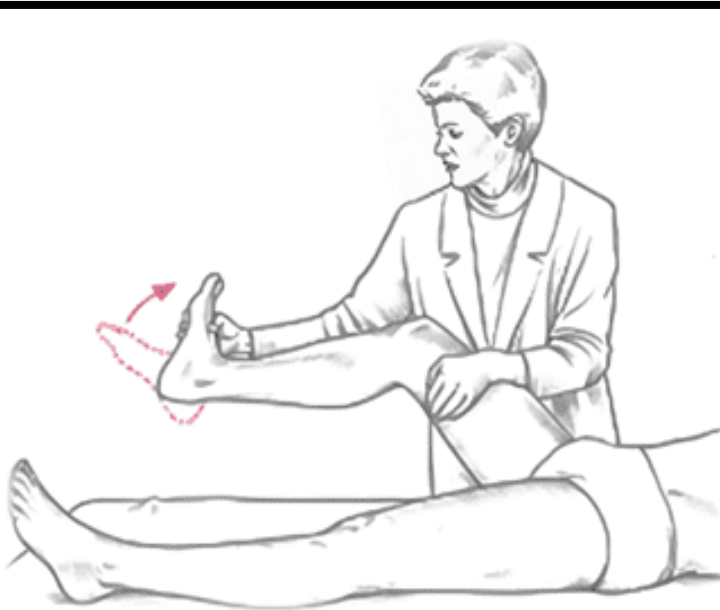
**Report any
abnormality**



The nurse assesses the patient's legs daily for calf tenderness, calf edema, and a positive Homan's sign



Homan's sign Definition: Pain, tenderness in the calf or behind the knee when the foot is **dorsiflexed** while the knee is **slightly flexed**



Homan's Sign

2- Nursing Diagnoses

1. **Pain** related to the surgical procedure.
2. **Impaired physical mobility** related to pain, swelling.
3. **Altered body image and self-esteem** related to impact of musculoskeletal problems.
4. High risk **for altered peripheral tissue perfusion** related to swelling and impaired circulation.
5. High risk **for fluid volume deficit** related to excessive losses from surgery.
6. High risk **for infection** related to traumatized tissues.

3- Planning

The major goals of the patient after orthopedic surgery may include:

1. Relief from pain.
2. Improve mobility
3. Adequate tissue perfusion
4. Health maintenance
5. Improved mobility
6. Improved self-concept
7. Absence of complications



4- Implementation

Pain related to the surgical procedure

Nursing Interventions:

1. **Analgesic** as prescribed to control the pain (used as a preventive).
2. **Elevation** of the operative extremity and application of **cold compresses** , if prescribed, help to control edema.
3. **Portable suction of the wound** decreases fluid accumulation and hematoma formation.
4. After 3 to 4 days, most patients require only oral analgesia.



Impaired physical mobility related to pain, swelling

Nursing Interventions:

- 1- Some orthopedic procedures require prolonged protection from excessive movement, weight-bearing restrictions, immobilizing splints, casts.
- 2-Using of **walking aid** (crutches, walker) which facilitate postoperative mobility.
- 3- Preoperative practice with mobility aids helps the patient use them postoperatively.
- 4- **Rehabilitation** involves progressively increasing the patient's activities and exercises as prescribed.



High risk for fluid volume deficit related to excessive losses from surgery

Nursing Interventions:

1- Identifies **early signs** and **symptoms** of shock and reports the findings to the orthopedic surgeon

2- Frequent monitoring of vital signs and report any abnormalities

3- Frequent monitoring of intake and output.

4- Fluid replacement accordingly.

01. Rapid Heartbeat

02. Low Blood Pressure

03. Confusion or Anxiety

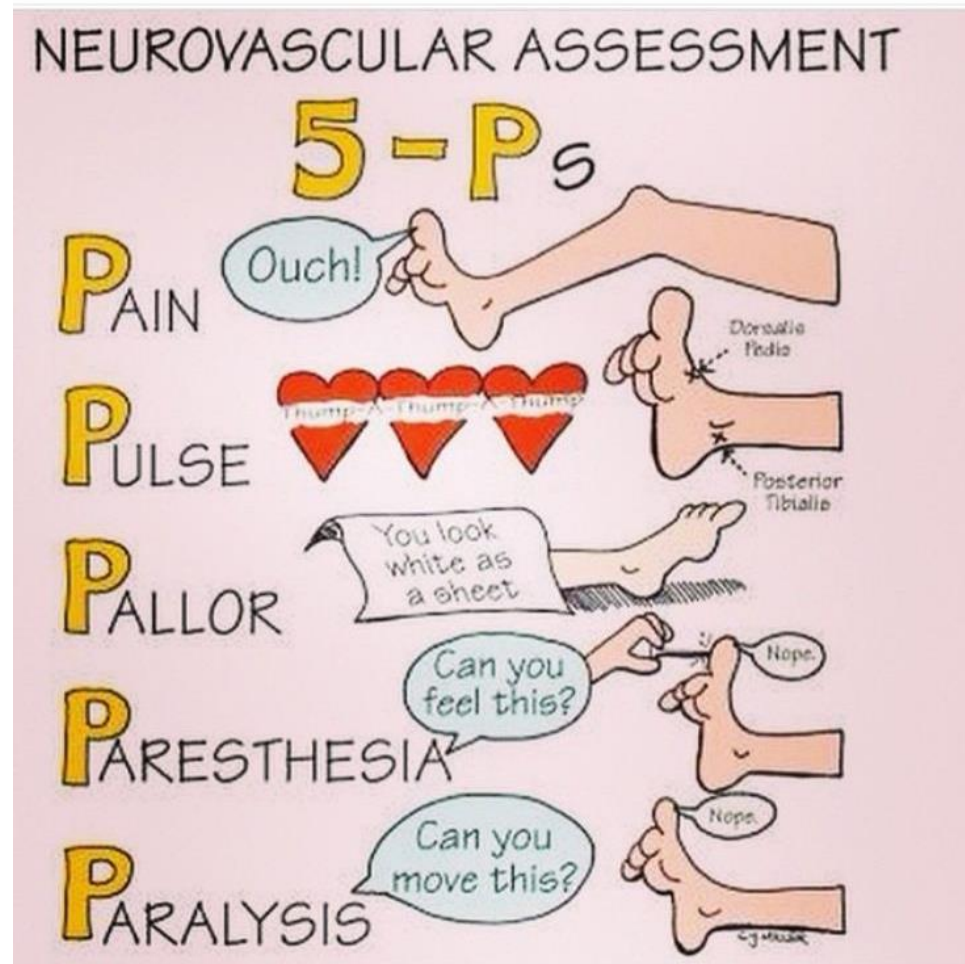
04. Cold Clammy Skin

05. Feeling Weak & Tired

High risk for altered peripheral tissue perfusion related to swelling and impaired circulation

Nursing Interventions:

The nurse monitors neurovascular status of the involved part.(5 P)



The nurse should remind the patient to **ankle and calf pumping** exercises as prescribed to enhance circulation.

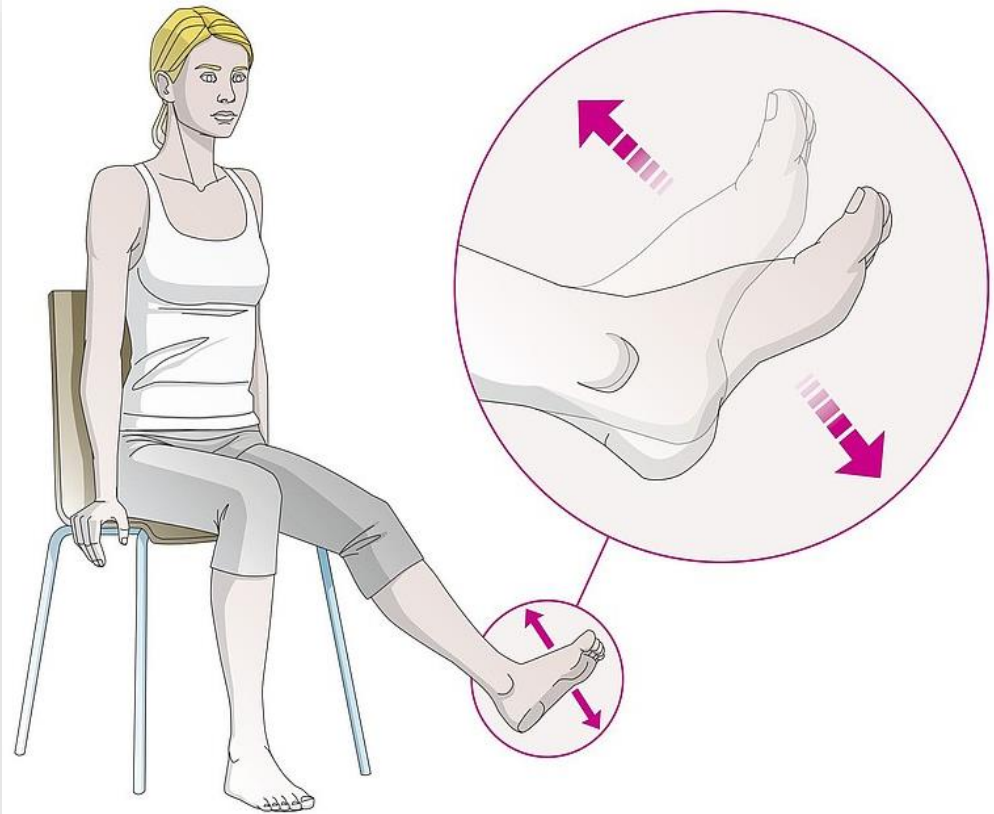
PROPER ANKLE PUMP EXERCISE



PLANTAR FLEXION



DORSIFLEXION



High risk for infection related to traumatized tissues

Nursing Interventions:

1. Prophylactic systemic antibiotics are frequently prescribed during the preoperative and immediate postoperative period.
2. The nurse assesses the patient's response to these antibiotics.
3. Following aseptic technique in wound dressing is essential.
4. The nurse monitors the patient's vital signs, assesses the wound, and notes the character of the drainage.
5. Hand washing, and frequent wound care reduces the development of infections.

5- Evaluation



Expected outcomes: The patient

1. Reports controlled pain.
2. Exhibits adequate tissue perfusion
3. Maximizes mobility within the therapeutic limits
4. Expresses positive self-concept.
5. Promotes health
6. Exhibits absence of complications





Summary



- List **3 types** of hip replacement
- List **4 nursing diagnosis** for patient undergoing orthopedic surgery
- List **5ps** for neurovascular assessment



Thank
you